

Latest Palm OS® System: Overview

David Fedor
Senior Developer Support Engineer

3Com
More connected.

Overview

PalmSource 99

- Palm OS® system 3.5
- Still in development
- Platform release; not tied to a particular device
- Flashable OS update for 3.0+ devices
  - Each device manufacturer controls timing and method of the release to their customers
  - Not all devices are flashable
- When

Overview (Cont.)

PalmSource 99

- New/improved
  - User-oriented features
  - Licensee features
  - Developer features
- Changes to built-in apps but not data
  - (except for the Launcher)
- New APIs
- Minimal desktop software changes

Product Objectives

PalmSource 99

- Maintain Platform market share
- Grow the Handheld market
- Create a stable and evolving platform
- Demonstrate that the Palm Computing® platform is the most appropriate choice for targeted device types

Technical Benefits

PalmSource 99

- Grand unification of Palm OS® systems 3.1, 3.2, 3.3, and new material
- Add and improve OS items that aren't specifically device-oriented
- Obvious benefits to users, to encourage upgrades, which benefits developers

Sessions Overview

PalmSource 99

- User-visible OS and application changes
  - This session
- Graphics system
  - Session 208, next
- UI Widgets, Notification Mgr, Other APIs
  - Session 210, next after 208
- Overlay Manager
  - Session 216, tomorrow
- Development Tools
  - This session, the others, and more


PalmSource™ 99
Demo
3Com
More connected.

Graphics System

- Support for 1-, 2-, 4-, and 8-bit color
- Support for 4-bit gray on EZ-based devices
- New Graphics APIs and developer features
- Greater consistency and completeness
- Access to change the UI element colors on a system-wide basis

Built-in App Changes

PalmSource 99

- Agenda View
- Alarm dialog
- Beaming categories
- Auto-completion in some fields
- Duplicating address book records
- “Selected” icon for Launcher
- No changes to PIM database records (yet)

Masked Records

PalmSource 99

- A record is still either private or public; the application display is what changes
- APIs to switch modes easily
- Tables have built-in support to do most of the work

Command Bar

- Contents under control of the app
  - Based on the user's context
- You add buttons when you see an event
- Typically these mirror menu commands, so no extra code necessary
- Fields will add cut/copy/paste/undo commands automatically
  - (Unless you tell them not to)

Misc. UI Changes

- Tapping title tab brings down menus
- Double/triple-tapping words in fields
- Category editing indication changed
  - Popup arrow if you're changing the app's view
  - Dotted rectangle if tapping there changes where the record is filed

SDK and Tools Changes

PalmSource 99

- Headers unified, clarified, improved, but apps still compatible (backward and forward)
- New headers have filename and data type changes for consistency and clarity
  - #include <PalmOS.h>
  - Int16, UInt16 instead of Int and Word, etc.
  - MemHandle & MemPtr, not Handle & Ptr
- Tools changes to support new UI widgets, color bitmaps, Overlays, etc.
- Test with rom on Palm OS® Emulator

API Changes

PalmSource 99

- Plenty of good new functions to support the new UI features
- Functions added to save time and/or application size and effort, for completeness, consistency, etc.
- Lower-level function changes
- Come to the talk later for full details

Release vs. Debug roms

PalmSource 99

- Using the debug rom in the emulator catches application bugs galore
- The debug rom keeps getting more checks
  - Parameter validation
  - Form object validation
- Many ErrFatalDisplay calls are now properly NonFatal, so you'll see them but not your customers
  - On debug rom, hit "Cont" to see if your app handles the error code properly
- New specific code for debugging

Compatibility

- Well-behaved applications should be fine
- Areas to particularly watch:
  - Directly accessing graphics globals, buffers, chip modes, etc. (Call ScrDisplayMode!)
  - Using NULL window and/or form pointers
  - Drawing to the screen at the wrong time
  - Assuming A4 register isn't touched
  - Card 1, and it possibly being write-protected
  - Ignoring error return codes
  - …and many things the debug rom will catch
- Visit the Compatibility talk (Friday)

Tools

- Palm OS® Emulator
  - Color, grays, Gremlin Hordes, etc.
- PalmRez
  - Color, grays, UI widgets, bitmap families
- Constructor
  - Color, new controls, direct PRC editing
- Simulator (Mac only)
  - Color
- The 3.5 SDK

What To Do Now

PalmSource 99

- Visit the appropriate talks and labs
- Get the CD tonight
- Try your app on the new OS
- Implement new features
- Give feedback
- Join the Development Services Seeding program (to get newer rom images)

PalmSource™ 99

Q&A

3Com
More connected.


PalmSource™ 99
3Com
More connected.